

Functional Annotation Report: PqqD2 (Q88JG8 / PP_2681) in *Pseudomonas putida* KT2440

Summary

PqqD2 (UniProt **Q88JG8**, gene *pqqD2*, ordered locus **PP_2681**) is a small (90-amino-acid) **peptide-binding chaperone** of the **PqqD** family — the founding member of the **RiPP precursor peptide Recognition Element (RRE)** superfamily. Its primary molecular function is **not enzymatic**. Rather, it acts as a **structural adapter**: it binds the ribosomally synthesized precursor peptide **PqqA** and presents it to the radical S-adenosylmethionine (radical SAM) enzyme **PqqE**, thereby initiating the biosynthesis of **pyrroloquinoline quinone (PQQ)**, a peptide-derived redox cofactor. PqqA is intrinsically unstructured in solution and can only productively engage PqqE when it is held in complex with a PqqD chaperone. This function is performed in the **cytoplasm**, where PQQ biosynthesis occurs. The gene symbol is **not ambiguous**: UniProt, InterPro, Pfam, STRING, and the primary structural literature all converge on the same RRE-chaperone function.

P. putida KT2440 is unusual in encoding **two PqqD paralogs**. The first, **PqqD1** (PP_0377), lies within the canonical *pqqFABCDEG* biosynthetic operon. The second, **PqqD2** (PP_2681) — the subject of this report — is located far from that operon and is instead **embedded in the "Ped" periplasmic alcohol/aldehyde-oxidation gene cluster**, immediately adjacent to the PQQ-dependent alcohol dehydrogenase genes *pedH* and *pedE*. Because the genome contains only a **single PqqA precursor gene** (PP_0380), both PqqD paralogs must act on the same substrate peptide. This genomic context, together with high-confidence functional-association networks, indicates that PqqD2 is a **redundant / co-regulated back-up chaperone** whose activity supplies PQQ specifically to the periplasmic dehydrogenase system (PedH/PedE, and glucose dehydrogenase) that this cluster serves.

In short: PqqD2 is a cytoplasmic RRE peptide chaperone that channels the PqqA precursor into PQQ biosynthesis. The PQQ cofactor it helps produce is then exported and used by periplasmic quinoprotein dehydrogenases. Its defining feature relative to its paralog is genomic and regulatory — it is co-located with, and likely dedicated to, the lanthanide/calcium-switchable Ped alcohol-oxidation module.

Key Findings

Finding 1 — PqqD2 is a PqqD-family RRE peptide chaperone for PQQ biosynthesis

PqqD2 is a 90-residue protein carrying the complete diagnostic set of PqqD/RRE domain signatures: InterPro **IPR008792 (PQQD)**, **IPR022479 (PqqD_bac)**, **IPR041881 (PqqD_sf)**, and Pfam **PF05402**. UniProt annotates its molecular function directly: *"Functions as a PqqA binding protein and presents PqqA to PqqE."* Its quaternary state is a monomer that interacts with PqqE, and it is assigned to the pyrroloquinoline quinone biosynthesis pathway.

PqqD is the **founding, prototypical member of the RRE (RiPP precursor peptide Recognition Element) family** — a class of small domains that recognize the leader/precursor peptides of ribosomally synthesized and post-translationally modified peptides (RiPPs) and deliver them to their tailoring enzymes. The mechanistic definition is well established in the primary literature: biosynthesis of PQQ *"is initiated when the precursor peptide, PqqA, is recognized and bound by the RiPP precursor peptide recognition element (RRE), PqqD, for presentation to the first enzyme in the pathway, PqqE"* (PMID: 28481092). Critically, the chaperone role is obligatory because the substrate is disordered on its own: *"PqqA is unstructured in solution, and only binds to PqqE when in complex with PqqD"* (PMID: 27638737).

This establishes PqqD2's function as a **molecular adapter / peptide presentation factor**, not a catalyst. It does not itself modify PqqA; it holds and orients the precursor so the downstream radical SAM enzyme can act. The "substrate" of PqqD2 is therefore a peptide (PqqA), not a small molecule.

Finding 2 — *P. putida* KT2440 encodes two PqqD paralogs (~48% identical)

Genome mapping of *P. putida* KT2440 (taxid 160488) shows the canonical *pqq* operon spanning PP_0381 (*pqqF*) – PP_0380 (*pqqA*) – PP_0379 (*pqqB*) – PP_0378 (*pqqC*) – PP_0377 (**pqqD1**, Q88QV7, 91 aa) – PP_0376 (*pqqE*). The second PqqD paralog, **pqqD2** (Q88JG8, PP_2681, 90 aa), lies far from this cluster.

A global Needleman–Wunsch alignment of the two proteins gives **42/87 aligned positions identical (48.3% identity)**, with the functionally critical C-terminal RRE core (the ... FMEVARA...WI motif region) conserved between them. Both paralogs carry an identical InterPro PqqD signature set (IPR008792 / IPR022479 / IPR041881). The moderate (~48%) sequence divergence combined with fully conserved domain architecture is consistent with an **ancient duplication that has retained the core peptide-recognition function while diverging in regulatory/contextual role** — a hallmark of a specialized paralog rather than a pseudogene.

Feature	PqqD1	PqqD2
Locus tag	PP_0377	PP_2681
UniProt	Q88QV7	Q88JG8
Length (aa)	91	90
Genomic context	Canonical <i>pqqFABCDEG</i> operon	Ped alcohol-oxidation cluster
Domain signatures	IPR008792/022479/041881	IPR008792/022479/041881
Pairwise identity	48.3% (42/87)	48.3% (42/87)

Finding 3 — PqqD acts within the cytoplasmic PQQ biosynthetic pathway; the PQQ product serves periplasmic dehydrogenases

PQQ is a **peptide-derived redox cofactor** produced from the RiPP precursor PqqA by a conserved set of enzymes, PqqB–E. The pathway logic is well characterized: PQQ is "*produced from a ribosomally synthesized and post-translationally modified peptide PqqA via a pathway comprising four conserved proteins PqqB-E. These four proteins are now fairly well-characterized and span radical SAM activity (PqqE), aided by a peptide chaperone (PqqD), a dual hydroxylase (PqqB), and an eight-electron, eight-proton oxidase (PqqC)*" (PMID: 31427437). This places PqqD's chaperone action at the **very first committed step** of the pathway, upstream of the radical-SAM-generated carbon–carbon cross-link between the conserved glutamate and tyrosine of

PqqA catalyzed by PqqE. Downstream, a protease excises the cross-linked intermediate, and PqqB (a dual hydroxylase) and PqqC (an eight-electron/eight-proton oxidase/cyclase) complete PQQ maturation (PMID: 32731194).

The subcellular logic of PQQ metabolism is **two-compartment**: biosynthesis (including the PqqD-mediated presentation step) occurs in the **cytoplasm**, whereas the finished PQQ cofactor is used by **periplasmic** quinoprotein dehydrogenases. In *P. putida* KT2440, PQQ is the redox coenzyme of a periplasmic glucose dehydrogenase: glucose is oxidized to gluconate "by a periplasmic glucose dehydrogenase (GDH) that requires pyrroloquinoline quinone (PQQ) as a redox coenzyme" (PMID: 27287323). This gluconic-acid-producing activity underlies phosphate solubilization in the rhizosphere, connecting PqqD-initiated cofactor supply to an ecologically important physiological output.

Finding 4 — pqqD2 is embedded in the Ped periplasmic alcohol/aldehyde-oxidation cluster, not the canonical pqq operon

The defining distinction of PqqD2 from its paralog is its genomic neighborhood. Around PP_2681, the gene order is: PP_2679 **pedH** (Q88JH0, 595 aa, quinoprotein / lanthanide-dependent PQQ alcohol dehydrogenase); PP_2680 **aldB-II** (aldehyde dehydrogenase); PP_2681 **pqqD2** (Q88JG8); PP_2682 **yiaY** (iron alcohol dehydrogenase). This module is flanked by two-component sensor histidine kinase genes (PP_2664 and PP_2683 / *yiaZ*) and downstream peptidase/transglutaminase-domain genes (PP_2685 / PP_2686). This locus is remote from the canonical *pqq* operon (PP_0376–PP_0381) that houses *pqqD1*.

The neighboring alcohol dehydrogenase PedH was characterized as "*the first description and characterization of a lanthanide-dependent PQQ-ADH (PedH)*" and this class of enzymes is part of periplasmic oxidation systems: "*many Gram-negative bacteria have evolved periplasmic oxidation systems based on pyrroloquinoline quinone-dependent alcohol dehydrogenases (PQQ-ADHs) that are often functionally redundant*" (PMID: 28655819). The embedding of pqqD2 within this cluster strongly suggests that this PqqD paralog is **co-regulated with the periplasmic alcohol-oxidation machinery it ultimately supplies with cofactor** — i.e., it is a locally deployed source of the PQQ-biosynthesis initiation function tied to Ped-system expression.

Finding 5 — STRING network links PqqD2 to both PQQ biosynthesis enzymes and PQQ-dependent alcohol dehydrogenases

Computational functional-association analysis (STRING v12, organism 160488; PP_2681 labeled "pqqD-II") confirms the dual connectivity implied by genomic context. PqqD2 shows **high-confidence associations to the core PQQ biosynthesis enzymes** — pqqD-II-pqqC 0.916, pqqD-II-pqqB 0.914, pqqD-II-pqqE 0.875 — and simultaneously to the **alcohol/aldehyde-oxidation cluster** — pqqD-II-qedH-I 0.78, pqqD-II-qedH-II 0.822, pqqD-II-yiaY 0.925, pqqD-II-PP_2683 0.927, pqqD-II-aldB-II 0.69. As an internal consistency check, the biosynthetic enzymes pqqB/C/E interconnect at 0.996–0.998, as expected for a tightly coupled pathway.

Association (PP_2681 = "pqqD-II")	STRING score	Functional group
pqqD-II – pqqC	0.916	PQQ biosynthesis
pqqD-II – pqqB	0.914	PQQ biosynthesis
pqqD-II – pqqE	0.875	PQQ biosynthesis
pqqD-II – yiaY	0.925	Ped alcohol-oxidation cluster
pqqD-II – PP_2683 (yiaZ)	0.927	Ped cluster regulator (TCS)
pqqD-II – qedH-II	0.822	PQQ-dependent ADH
pqqD-II – qedH-I	0.78	PQQ-dependent ADH
pqqD-II – aldB-II	0.69	Ped aldehyde DH

This bipartite network is the strongest single piece of evidence that PqqD2 functionally bridges **cofactor manufacture** (PQQ biosynthesis) and **cofactor consumption** (periplasmic dehydrogenases) — consistent with a paralog specialized to serve the Ped system.

Finding 6 — A single PqqA precursor means both PqqD paralogs present the same substrate peptide

A proteome search of *P. putida* KT2440 returns **exactly one PqqA gene**, PP_0380 (Q88QV4, 23 aa); there is no *pqqA2* or second precursor, and the Ped cluster surrounding pqqD2 contains no dedicated PqqA-like ORF. This is a decisive constraint: whatever the two PqqD paralogs do, they must both act on the **same, shared 23-residue precursor peptide**. PqqD2 therefore is not a chaperone for a distinct RiPP; it is a redundant/specialized presenter of the canonical PqqA.

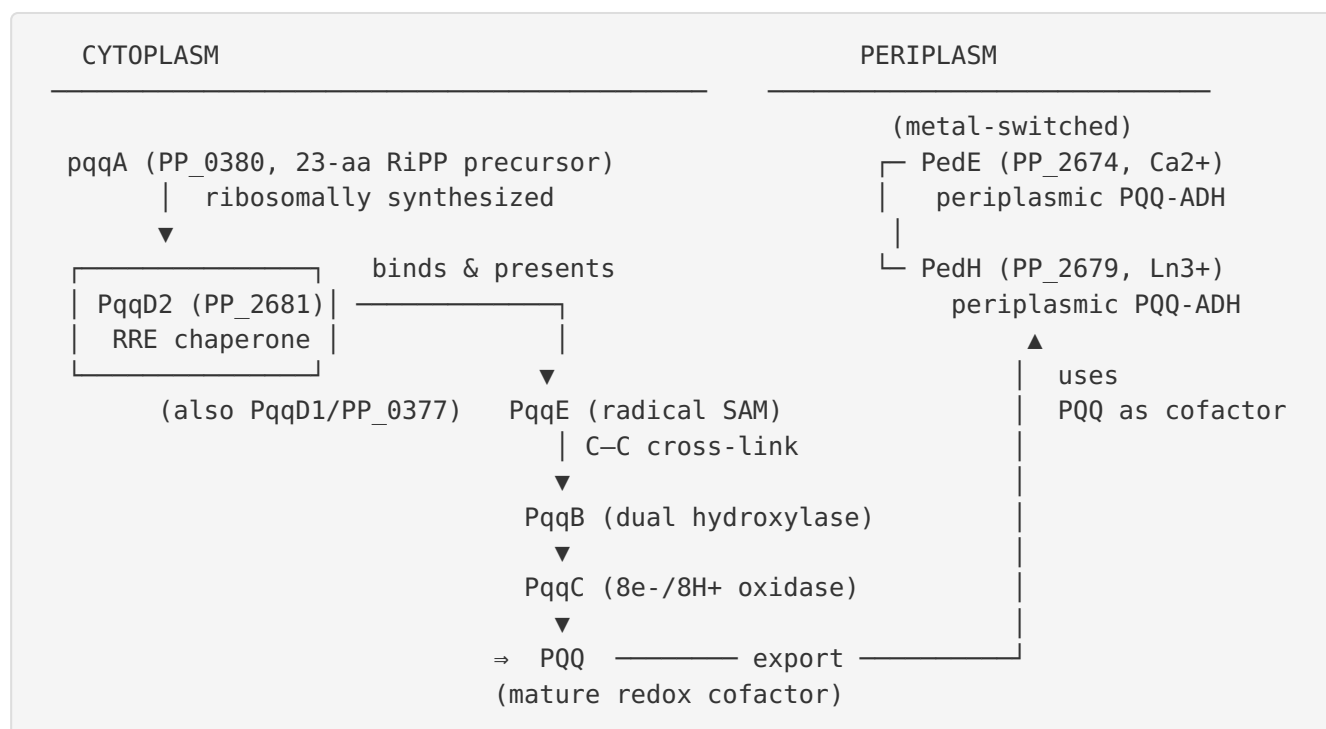
Structurally, the AlphaFold model of PqqD2 (Q88JG8) is high-confidence — mean pLDDT 92.4, with 92% of residues above 90 and 98% above 70 — indicating a **well-ordered, compact winged-helix/RRE domain** that is fully competent for peptide binding. PqqD2 is thus a folded, functional protein, not a degenerate remnant.

Finding 7 — The Ped cluster encodes periplasmic PQQ-dependent alcohol dehydrogenases PedE (Ca) and PedH (Ln), defining the downstream utilization compartment

The two paralogous PQQ-alcohol dehydrogenases served by this locus are **pedE** = **PP_2674** (Q88JH5, 631 aa) and **pedH** = **PP_2679** (Q88JH0, 595 aa). Both are annotated "Quinoprotein alcohol dehydrogenase," both carry an N-terminal signal peptide, and both have subcellular location = **periplasm**. PedH is **lanthanide-dependent** and PedE is **calcium-dependent** ([PMID: 28655819](#)); the two enzymes form a metal-responsive switch (the "lanthanide switch") governing periplasmic alcohol oxidation. Because pqqD2 (PP_2681) sits physically between these dehydrogenase genes and the aldehyde/alcohol DH genes, it is spatially and regulatorily embedded in the machinery whose catalytic activity absolutely depends on the PQQ cofactor that PqqD2 helps generate.

Mechanistic Model / Interpretation

The findings assemble into a coherent two-compartment model in which PqqD2 is the entry-point chaperone for a locally deployed copy of the PQQ biosynthesis initiation step.



Molecular function. PqqD2 is a **peptide-recognition adapter**, the prototypical RRE. It binds the disordered PqqA precursor and presents it to PqqE, enabling the first, committed catalytic step of PQQ biosynthesis. It has no catalytic activity of its own. Its oligomeric state is monomeric (physiologically), interacting with PqqE, and its substrate is the peptide PqqA rather than any small molecule.

Localization. The chaperone step is **cytoplasmic**, consistent with the location of all PQQ biosynthetic chemistry; PqqD2 has no signal peptide. The mature PQQ product is subsequently **exported to the periplasm**, where it is the essential redox cofactor of quinoprotein dehydrogenases — glucose dehydrogenase and, most relevantly for this locus, the alcohol dehydrogenases PedE and PedH.

Pathway role and the meaning of the paralog. The single most informative fact about PqqD2 is its genomic and network context. Unlike PqqD1, which is hard-wired into the housekeeping *pqqFABCDEG* operon, PqqD2 is embedded in the **Ped alcohol-oxidation cluster** and is functionally networked (STRING) to both the PQQ biosynthesis enzymes and the Ped dehydrogenases. Given that only one PqqA precursor exists, PqqD2 is best interpreted as a **redundant / back-up and locally co-regulated chaperone**: a second copy of the PqqA-presentation function positioned so that expression of the alcohol-oxidation system can be coupled to a dedicated supply of PQQ-biosynthesis initiation capacity. This is consistent with the general observation that Gram-negative periplasmic PQQ-ADH systems are "often

functionally redundant" ([PMID: 28655819](#)). The flanking two-component sensor kinases (PP_2664, PP_2683/yiaZ) provide a plausible regulatory route by which Ped-system induction (e.g., in response to alcohols or to lanthanide availability) could co-induce pqqD2.

The upshot: PqqD2 is not a novel enzyme or transporter. It is a well-folded structural adapter whose importance lies in *where* it is deployed — as the peptide-presentation entry point of PQQ biosynthesis, apparently dedicated to feeding cofactor into the periplasmic Ped alcohol-oxidation module.

Evidence Base

PMID	Title (abbrev.)	How it supports the findings
28481092	<i>NMR structure and binding studies of PqqD</i>	Defines PqqD as the RRE that binds PqqA and presents it to PqqE (Finding 1).
27638737	<i>NMR assignments for M. extorquens PqqD and PqqD–PqqA complex</i>	Shows PqqA is unstructured alone and binds PqqE only when in complex with PqqD — establishes obligatory chaperone role and monomeric action (Finding 1).
31427437	<i>A two-component protease in... / PQQ pathway proteins</i>	Defines the PqqB–E pathway and PqqD as the peptide chaperone aiding radical SAM PqqE (Finding 3).
27287323	<i>Regulation of PQQ-dependent glucose dehydrogenase in P. putida KT2440</i>	Establishes PQQ as the coenzyme of a periplasmic dehydrogenase in this exact organism; maps <i>pqqFABCDEG</i> operon (Finding 3).
28655819	<i>Functional role of lanthanides in PQQ-ADHs (PedH)</i>	Characterizes PedH (neighbor of <i>pqqD2</i>) as the first lanthanide-dependent periplasmic PQQ-ADH; notes functional redundancy of such systems (Findings 4, 7).
32731194	<i>Biogenesis of the peptide-derived redox cofactor PQQ (review)</i>	Authoritative review of PQQ biogenesis mechanism, contextualizing PqqD within PqqE/PqqF-G/PqqB chemistry (Finding 3).
32345644	<i>Cellular response to lanthanum in P. putida KT2440</i>	Physiological context for lanthanide-responsive PQQ-ADH metabolism in the same strain (supporting Ped-system relevance).
40031112	<i>Coordination number/ionic radius and Ln-dependent ADH activity</i>	Mechanistic detail on Ln–PQQ cofactor complexes in bacterial alcohol dehydrogenases (downstream utilization).

Database and computational evidence used: UniProt Q88JG8 (function, subunit, pathway, domains); InterPro/Pfam domain signatures (IPR008792, IPR022479, IPR041881, PF05402); genome mapping of taxid 160488 (operon and cluster gene order); Needleman–Wunsch pairwise alignment (48.3% identity to PqqD1); STRING v12 functional-association network; AlphaFold structural model (mean pLDDT 92.4).

Coherence of the evidence. The experimental literature (NMR structures, mechanistic biochemistry) firmly establishes the *molecular function* of PqqD proteins generally, in organisms such as *Methylobacterium extorquens*, *Xanthomonas campestris*, and *Klebsiella*. The organism-specific claims about PqqD2 (paralogy, genomic context, single shared PqqA, network associations, structural competence) rest on high-quality database, comparative-genomics, and bioinformatic evidence rather than on direct biochemical characterization of Q88JG8 itself. No evidence contradicts the RRE-chaperone assignment; all lines converge.

Limitations and Knowledge Gaps

- 1. No direct biochemical characterization of Q88JG8.** The RRE-chaperone function is assigned to PqqD2 by strong family/domain homology and structural modeling, not by direct in vitro demonstration that this specific paralog binds PqqA and stimulates PqqE. The mechanistic NMR/biochemistry was performed on PqqD from other organisms, not on the *P. putida* PP_2681 protein.
 - 2. The functional distinction between PqqD1 and PqqD2 is inferred, not proven.** The hypothesis that PqqD2 is a redundant/back-up chaperone dedicated to the Ped system is supported by genomic context and STRING associations, but no gene-knockout or expression study cited here directly demonstrates (a) that a $\Delta pqqD2$ mutant has a phenotype specific to alcohol oxidation, or (b) that PqqD1 and PqqD2 are functionally interchangeable.
 - 3. Regulatory coupling is a plausible model, not a measured fact.** The flanking two-component systems (PP_2664, PP_2683) suggest co-regulation of pqqD2 with the Ped cluster, but no transcriptomic evidence linking pqqD2 expression to alcohol/lanthanide induction was established here.
 - 4. Quantitative cofactor flux is unknown.** It is not known whether PqqD2 contributes meaningfully to total PQQ output under any condition, or whether it is essentially silent unless the Ped system is induced.
 - 5. The single-PqqA constraint, while strong, was derived from proteome/genome annotation** and would ideally be confirmed against the most recent genome build to exclude a poorly annotated or cryptic second precursor.
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Proposed Follow-up Experiments / Actions

1. **In vitro reconstitution of PqqD2 activity.** Express and purify recombinant Q88JG8 and test (a) direct binding to the PP_0380 PqqA peptide (ITC / fluorescence anisotropy) and (b) its ability to stimulate PqqE-catalyzed C–C cross-linking, compared side-by-side with PqqD1. This would convert the homology-based assignment into direct evidence.
 2. **Genetic dissection of paralog roles.** Construct $\Delta pqqD1$, $\Delta pqqD2$, and $\Delta pqqD1\Delta pqqD2$ mutants and assay PQQ production and growth on PQQ-dependent substrates (glucose→gluconate; alcohols oxidized by PedE/PedH). Test whether $\Delta pqqD2$ has a phenotype selective for the Ped/alcohol-oxidation branch, and whether either paralog alone suffices for PQQ synthesis.
 3. **Cross-complementation.** Test whether *pqqD2* expressed from a plasmid rescues a $\Delta pqqD1$ mutant (and vice versa) to determine functional interchangeability.
 4. **Transcriptional profiling.** Measure *pqqD2* expression under conditions that induce the Ped system (growth on alcohols such as 2-phenylethanol; presence vs. absence of lanthanides) and test dependence on the flanking two-component sensor kinases (PP_2664, PP_2683). This would confirm or refute the proposed regulatory coupling.
 5. **Structural confirmation and complex modeling.** Solve or refine the PqqD2–PqqA complex (experimentally or via AlphaFold-Multimer) and compare the peptide-binding interface with PqqD1 to identify residues that might tune substrate affinity or specificity between paralogs.
 6. **Genome/precursor audit.** Re-scan the current KT2440 genome assembly for any additional PqqA-like ORFs to definitively confirm the single-precursor constraint underpinning the redundancy model.
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Conclusion

PqqD2 (Q88JG8 / PP_2681) is confidently annotated as a **cytoplasmic PqqD-family RRE peptide chaperone**: it binds the ribosomal precursor peptide PqqA and presents it to the radical SAM enzyme PqqE to initiate PQQ cofactor biosynthesis. It is a structural adapter, not an enzyme, and it acts on the genome's single shared PqqA precursor. Its distinguishing biological feature is contextual — it is the second of two PqqD paralogs, uniquely embedded in the Ped periplasmic alcohol-oxidation cluster and networked to both PQQ biosynthesis

enzymes and the periplasmic PQQ-dependent dehydrogenases (PedE/PedH), marking it as a redundant, likely co-regulated chaperone that supplies cofactor-biosynthesis initiation capacity to the periplasmic alcohol-oxidation system. Direct biochemical and genetic studies of this specific paralog remain the key outstanding gap.

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